

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/dri

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
filepath = '/content/drive/MyDrive/colab files/Invistico_Airline.csv'
df = pd.read_csv(filepath)
df.head()
```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0	0	0	2
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0	0	0	3
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0	0	0	3
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0	0	0	3
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0	0	0	3

5 rows × 22 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 129880 entries, 0 to 129879
Data columns (total 22 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   satisfaction                          129880 non-null object
 1   Customer Type                        129880 non-null object
 2   Age                                  129880 non-null int64
 3   Type of Travel                       129880 non-null object
 4   Class                                129880 non-null object
 5   Flight Distance                      129880 non-null int64
 6   Seat comfort                         129880 non-null int64
 7   Departure/Arrival time convenient    129880 non-null int64
 8   Food and drink                       129880 non-null int64
 9   Gate location                        129880 non-null int64
10   Inflight wifi service                129880 non-null int64
11   Inflight entertainment               129880 non-null int64
12   Online support                       129880 non-null int64
13   Ease of Online booking               129880 non-null int64
14   On-board service                    129880 non-null int64
15   Leg room service                    129880 non-null int64
16   Baggage handling                    129880 non-null int64
17   Checkin service                     129880 non-null int64
18   Cleanliness                         129880 non-null int64
19   Online boarding                      129880 non-null int64
20   Departure Delay in Minutes           129880 non-null int64
21   Arrival Delay in Minutes             129487 non-null float64
dtypes: float64(1), int64(17), object(4)
memory usage: 21.8+ MB
```

```
df.describe()
```

	Age	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	wif:
count	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	1298
mean	39.427957	1981.409055	2.838597	2.990645	2.851994	2.990422	
std	15.119360	1027.115606	1.392983	1.527224	1.443729	1.305970	
min	7.000000	50.000000	0.000000	0.000000	0.000000	0.000000	
25%	27.000000	1359.000000	2.000000	2.000000	2.000000	2.000000	
50%	40.000000	1925.000000	3.000000	3.000000	3.000000	3.000000	
75%	51.000000	2544.000000	4.000000	4.000000	4.000000	4.000000	
max	85.000000	6951.000000	5.000000	5.000000	5.000000	5.000000	

df.isnull().sum()	
	0
satisfaction	0
Customer Type	0
Age	0
Type of Travel	0
Class	0
Flight Distance	0
Seat comfort	0
Departure/Arrival time convenient	0
Food and drink	0
Gate location	0
Inflight wifi service	0
Inflight entertainment	0
Online support	0
Ease of Online booking	0
On-board service	0
Leg room service	0
Baggage handling	0
Checkin service	0
Cleanliness	0
Online boarding	0
Departure Delay in Minutes	0
Arrival Delay in Minutes	393
dtype: int64	

df.duplicated().sum()

np.int64(0)

df.shape

```
(129880, 22)
```

```
df.columns
```

```
Index(['satisfaction', 'Customer Type', 'Age', 'Type of Travel', 'Class',
      'Flight Distance', 'Seat comfort', 'Departure/Arrival time convenient',
      'Food and drink', 'Gate location', 'Inflight wifi service',
      'Inflight entertainment', 'Online support', 'Ease of Online booking',
      'On-board service', 'Leg room service', 'Baggage handling',
      'Checkin service', 'Cleanliness', 'Online boarding',
      'Departure Delay in Minutes', 'Arrival Delay in Minutes'],
      dtype='object')
```

```
df.columns = df.columns.str.lower()
```

```
df.columns
```

```
Index(['satisfaction', 'customer type', 'age', 'type of travel', 'class',
      'flight distance', 'seat comfort', 'departure/arrival time convenient',
      'food and drink', 'gate location', 'inflight wifi service',
      'inflight entertainment', 'online support', 'ease of online booking',
      'on-board service', 'leg room service', 'baggage handling',
      'checkin service', 'cleanliness', 'online boarding',
      'departure delay in minutes', 'arrival delay in minutes'],
      dtype='object')
```

```
df['satisfaction'].value_counts()
```

	count
satisfaction	
satisfied	71087
dissatisfied	58793

dtype: int64

```
df.columns = df.columns.str.replace(' ', '_')
```

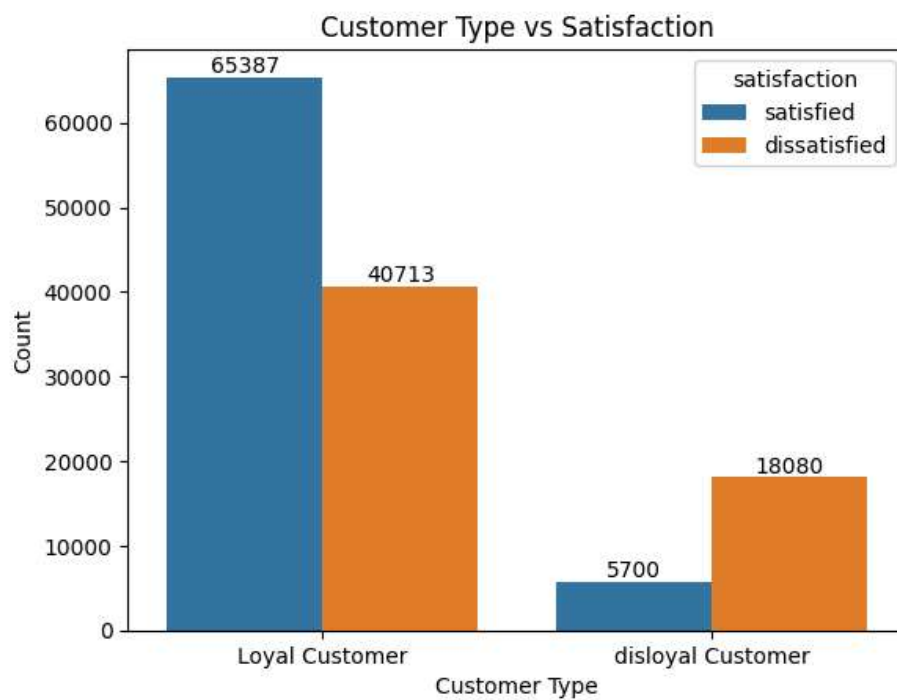
EDA

```
df.groupby('customer_type')['satisfaction'].value_counts()
```

		count
customer_type	satisfaction	
Loyal Customer	satisfied	65387
	dissatisfied	40713
disloyal Customer	dissatisfied	18080
	satisfied	5700

dtype: int64

```
ax = sns.countplot(x = 'customer_type', data = df, hue = 'satisfaction')
for containe in ax.containers:
    ax.bar_label(containe)
plt.title('Customer Type vs Satisfaction')
plt.xlabel('Customer Type')
plt.ylabel('Count')
plt.show()
```

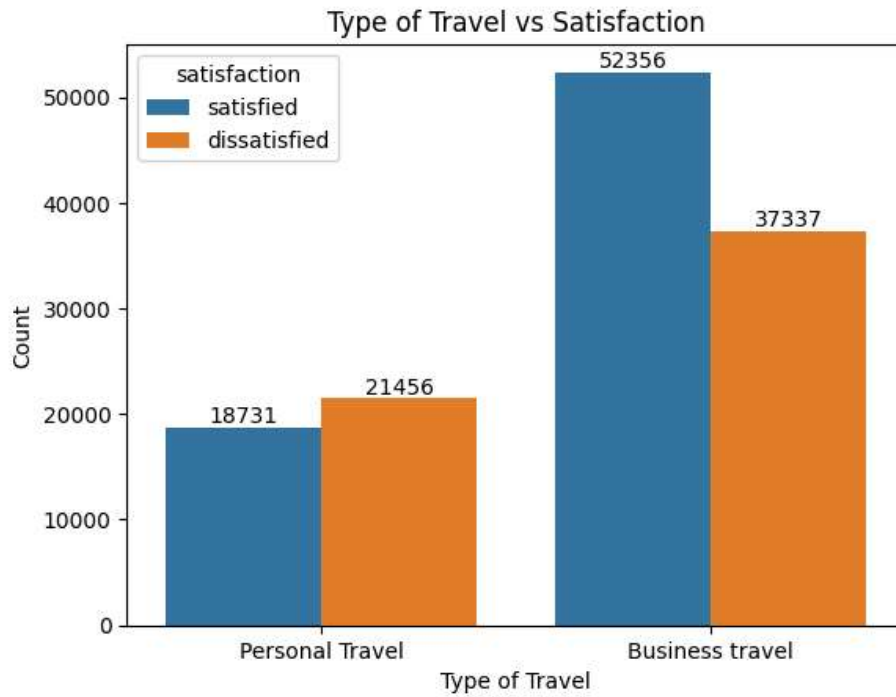


```
df.groupby('type_of_travel')['satisfaction'].value_counts()
```

		count
type_of_travel	satisfaction	
Business travel	satisfied	52356
	dissatisfied	37337
Personal Travel	dissatisfied	21456
	satisfied	18731

dtype: int64

```
ax = sns.countplot(x = 'type_of_travel', data = df, hue = 'satisfaction')
for container in ax.containers:
    ax.bar_label(container)
plt.title('Type of Travel vs Satisfaction')
plt.xlabel('Type of Travel')
plt.ylabel('Count')
plt.show()
```



```
df.groupby('class')['satisfaction'].value_counts()
```

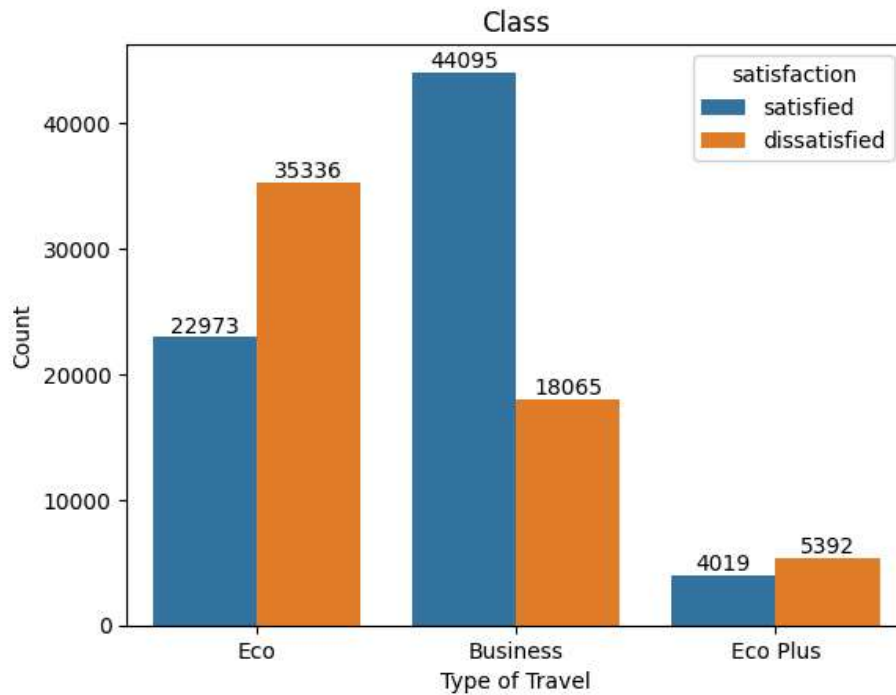
		count
class	satisfaction	
Business	satisfied	44095
	dissatisfied	18065
Eco	dissatisfied	35336
	satisfied	22973
Eco Plus	dissatisfied	5392
	satisfied	4019

dtype: int64

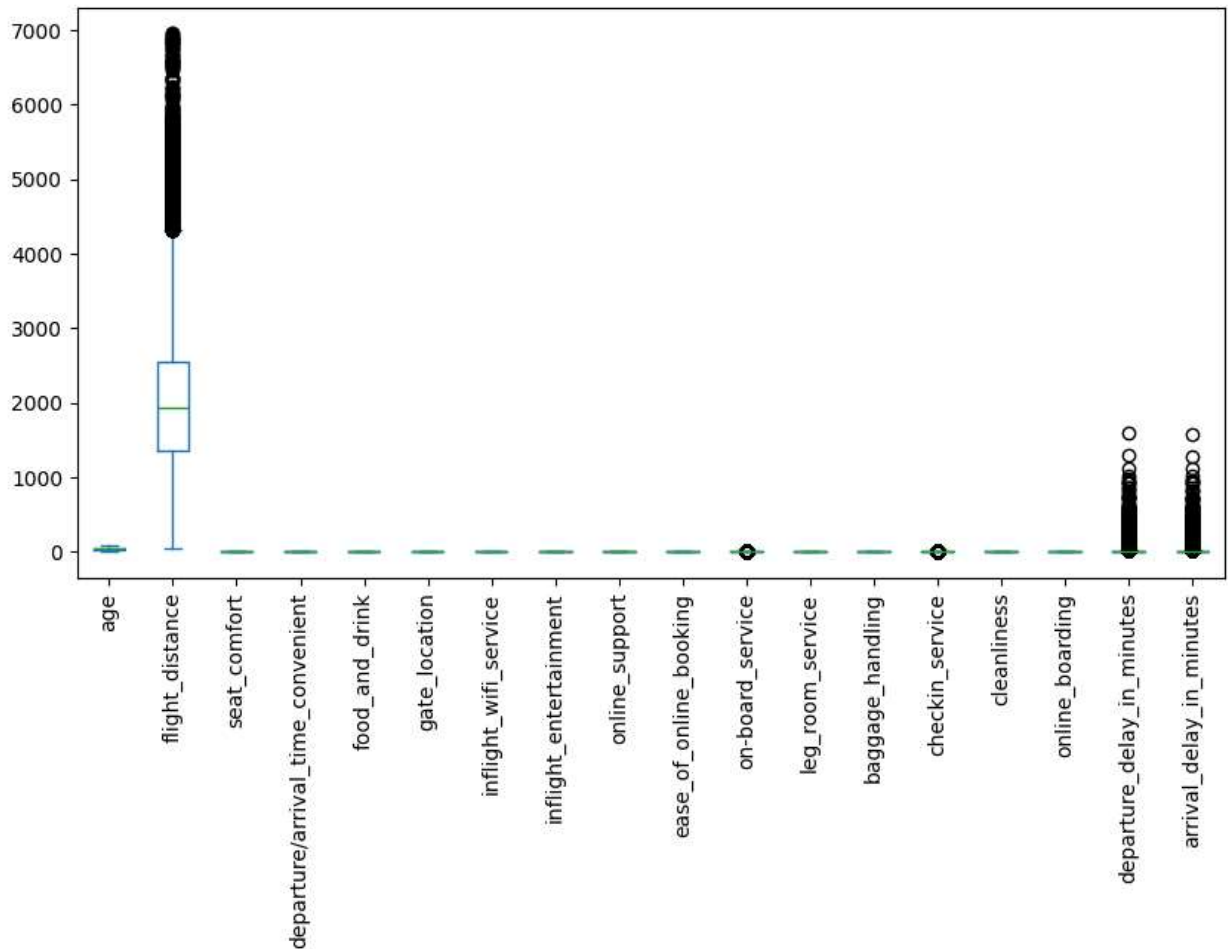
```
percentage = df.groupby('class')['satisfaction'].value_counts(normalize=True) * 100
percentage = percentage.reset_index(name='percentage')
percentage.sort_values(by='percentage', ascending=False)
```

	class	satisfaction	percentage
0	Business	satisfied	70.937902
2	Eco	dissatisfied	60.601279
4	Eco Plus	dissatisfied	57.294655
5	Eco Plus	satisfied	42.705345
3	Eco	satisfied	39.398721
1	Business	dissatisfied	29.062098

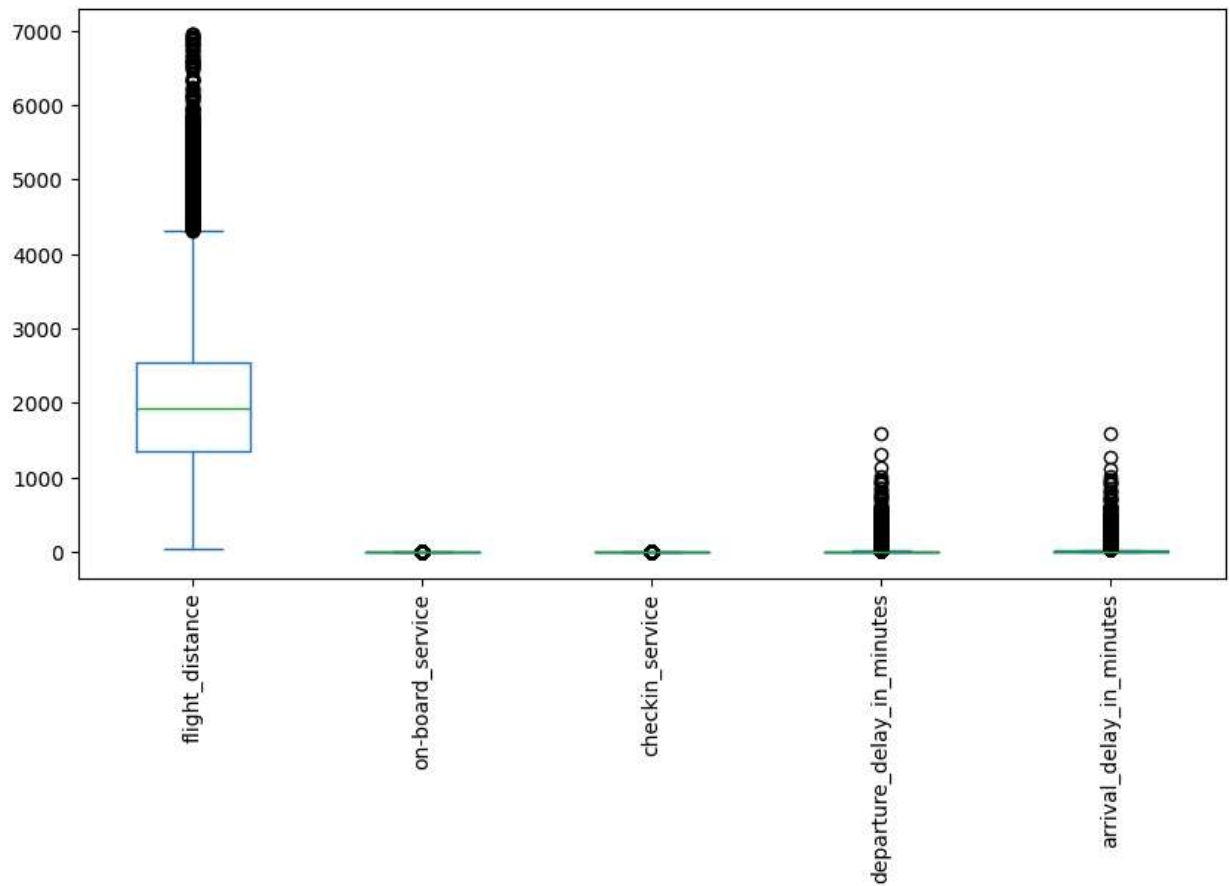
```
ax = sns.countplot(x = 'class', data = df, hue = 'satisfaction')
for container in ax.containers:
    ax.bar_label(container)
plt.title('Class')
plt.xlabel('Type of Travel')
plt.ylabel('Count')
plt.show()
```



```
df.plot(kind = 'box', figsize = (10, 5))  
plt.xticks(rotation = 90)  
plt.show()
```

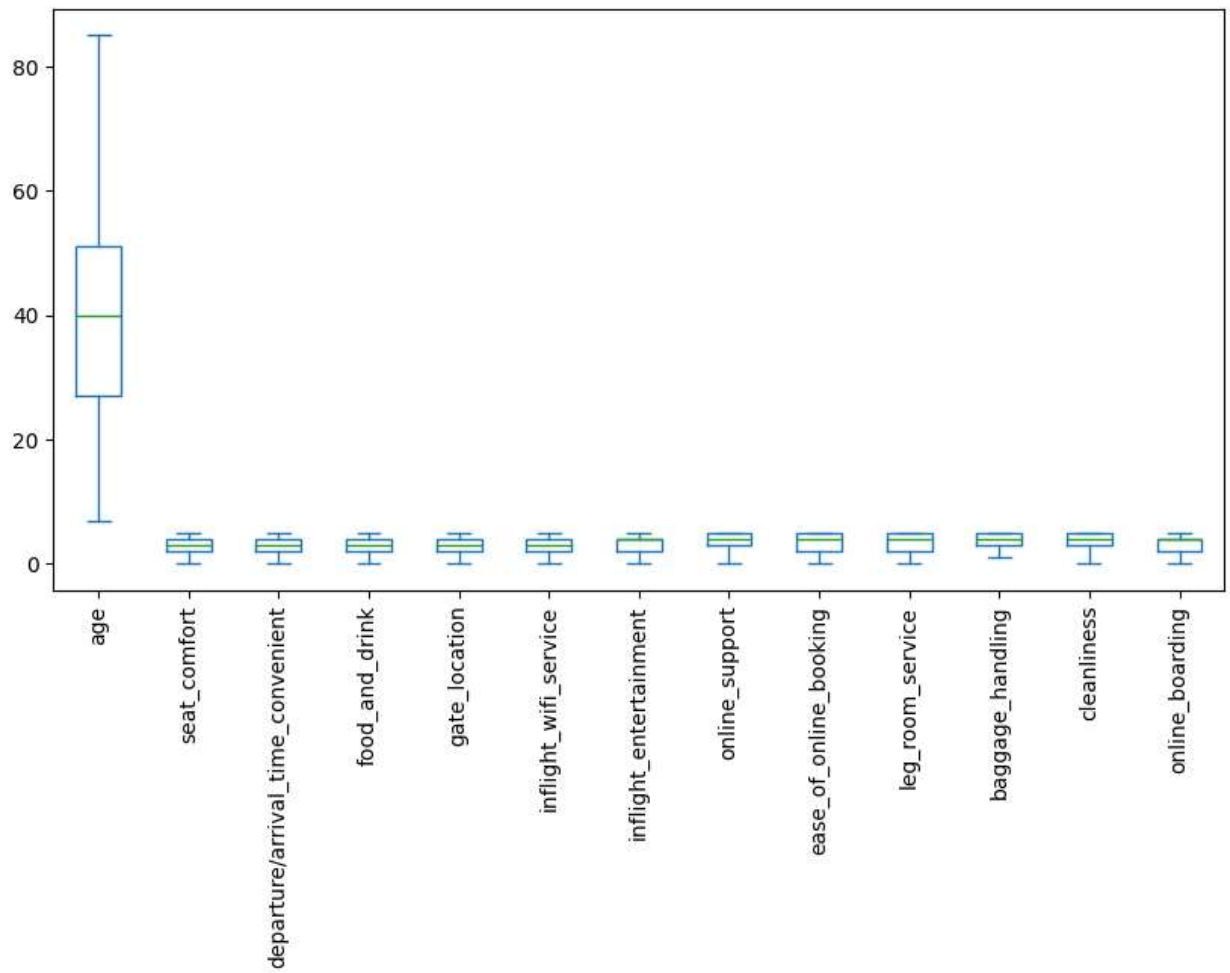


```
outl = df[['flight_distance', 'on-board_service', 'checkin_service', 'departure_delay_in_minutes', 'arrival_delay_in_minutes']]  
outl.plot(kind = 'box', figsize = (10, 5))  
plt.xticks(rotation = 90)  
plt.show()
```

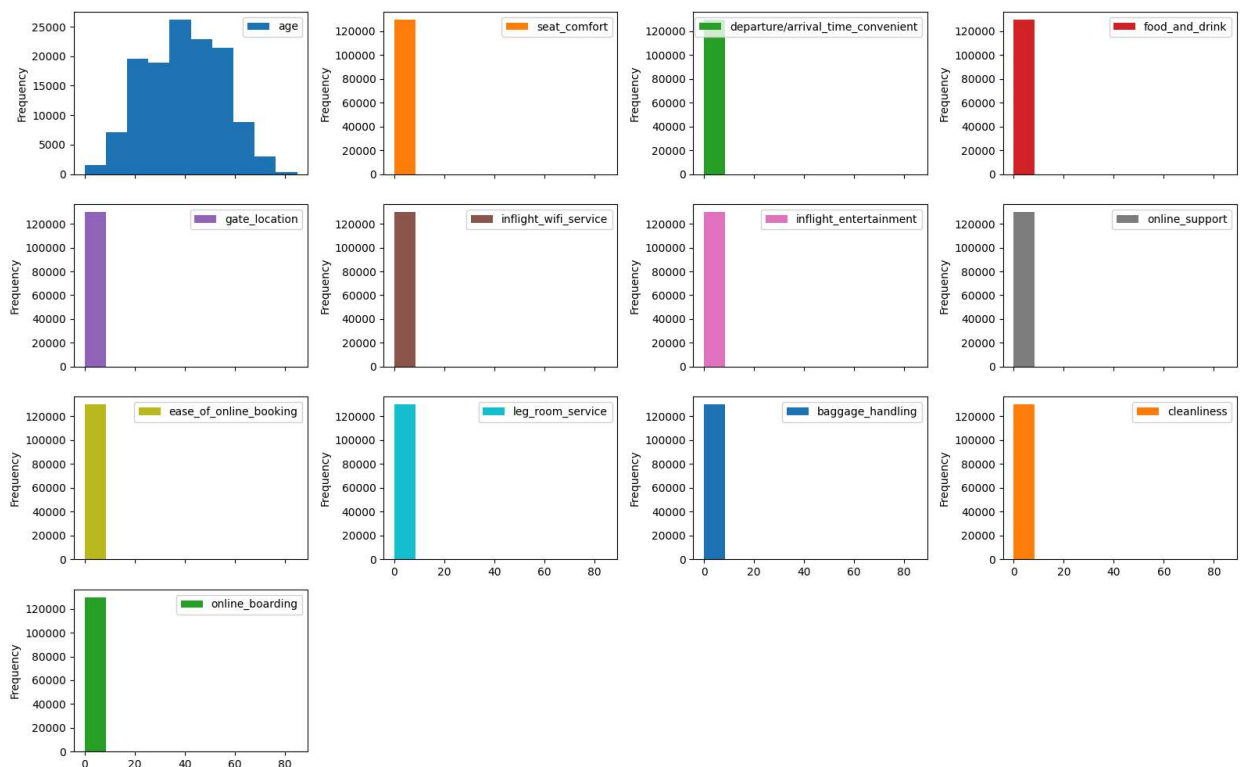


```
others = df[['age', 'seat_comfort', 'departure/arrival_time_convenient',  
            'food_and_drink', 'gate_location', 'inflight_wifi_service',  
            'inflight_entertainment', 'online_support', 'ease_of_online_booking',  
            'leg_room_service', 'baggage_handling',  
            'cleanliness', 'online_boarding']]
```

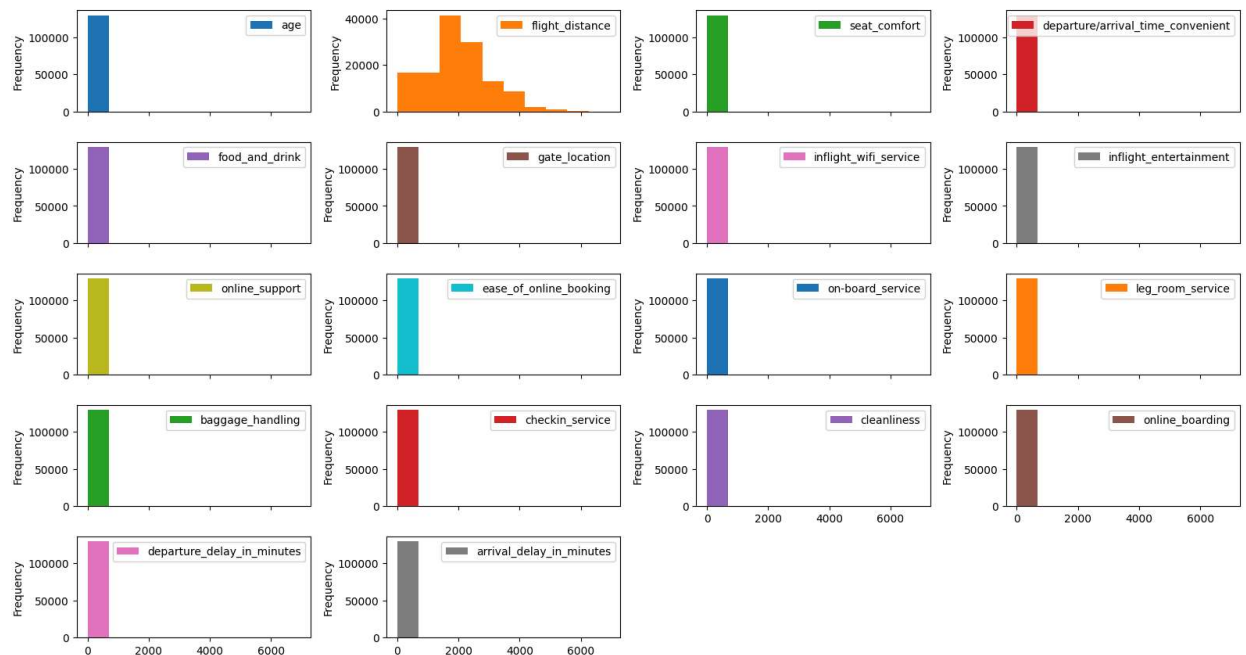
```
others.plot(kind = 'box', figsize = (10, 5))  
plt.xticks(rotation = 90)  
plt.show()
```



```
others.plot(kind = 'hist', subplots = True, figsize = (16, 10), layout = (4, 4))
plt.tight_layout()
plt.show()
```




```
df.plot(kind = 'hist', subplots = True, figsize = (16, 10), layout = (6, 4))
plt.tight_layout()
plt.show()
```



```
sat_col_cat = {'satisfied' : '1', "dissatisfied" : '0'}
df['satisfaction'] = df['satisfaction'].map(sat_col_cat)
```

```
df.head()
```

	satisfaction	customer_type	age	type_of_travel	class	flight_distance	seat_comfort	departure/
0	1	Loyal Customer	65	Personal Travel	Eco	265	0	
1	1	Loyal Customer	47	Personal Travel	Business	2464	0	
2	1	Loyal Customer	15	Personal Travel	Eco	2138	0	
3	1	Loyal Customer	60	Personal Travel	Eco	623	0	
4	1	Loyal Customer	70	Personal Travel	Eco	354	0	

5 rows × 22 columns

```
df.head()
```

	satisfaction	customer_type	age	type_of_travel	class	flight_distance	seat_comfort	departure/
0	1	Loyal Customer	65	Personal Travel	Eco	265	0	
1	1	Loyal Customer	47	Personal Travel	Business	2464	0	
2	1	Loyal Customer	15	Personal Travel	Eco	2138	0	
3	1	Loyal Customer	60	Personal Travel	Eco	623	0	
4	1	Loyal Customer	70	Personal Travel	Eco	354	0	

5 rows × 22 columns

```
df['arrival_delay_in_minutes'].agg(['mean', 'median', 'sum'])
```

```
arrival_delay_in_minutes
```

mean	1.509113e+01
median	0.000000e+00
sum	1.954105e+06

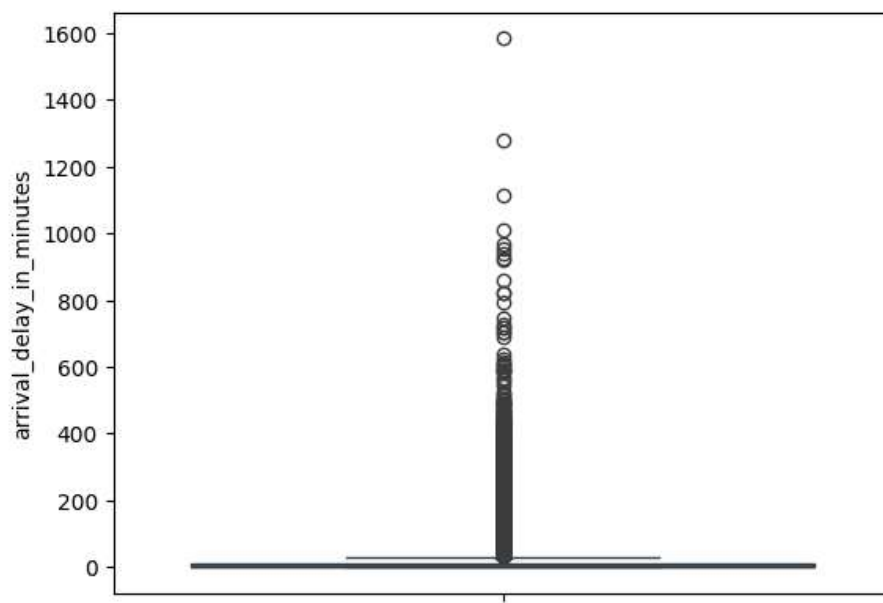
dtype: float64

```
df['arrival_delay_in_minutes'] = df['arrival_delay_in_minutes'].fillna(df['arrival_delay_in_minutes']).
```

```
cat_col = ['customer_type', 'type_of_travel', 'class']  
df = pd.get_dummies(df, columns = cat_col, drop_first = True, dtype = 'int')
```

```
sns.boxplot(y = 'arrival_delay_in_minutes', data = df)
```

<Axes: ylabel='arrival_delay_in_minutes'>



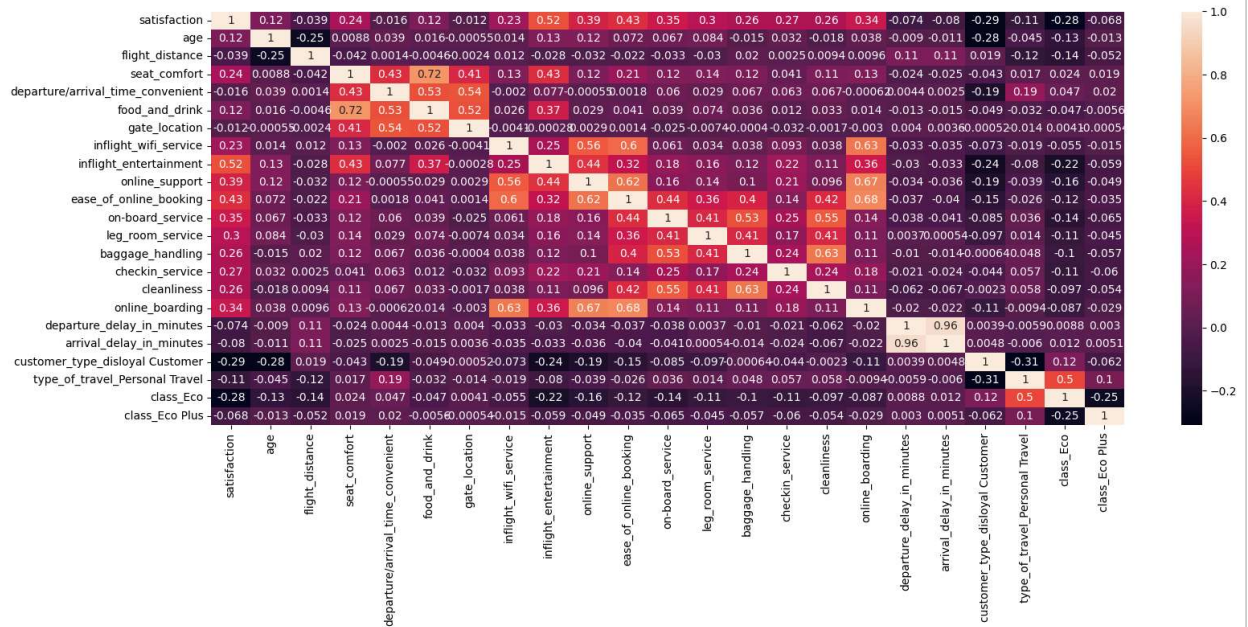
```
corr_mat = df.corr()  
corr_mat
```

	satisfaction	age	flight_distance	seat_comfort	departure/arrival
satisfaction	1.000000	0.117971	-0.039224	0.242384	
age	0.117971	1.000000	-0.249625	0.008784	
flight_distance	-0.039224	-0.249625	1.000000	-0.042409	
seat_comfort	0.242384	0.008784	-0.042409	1.000000	
departure/arrival_time_convenient	-0.015507	0.039055	0.001350	0.434951	
food_and_drink	0.120677	0.015749	-0.004649	0.715997	
gate_location	-0.012071	-0.000547	-0.002409	0.405300	
inflight_wifi_service	0.227062	0.013965	0.012230	0.129131	
inflight_entertainment	0.523496	0.130511	-0.027542	0.426447	
online_support	0.390143	0.121201	-0.032022	0.120278	
ease_of_online_booking	0.431772	0.071594	-0.022299	0.211531	
on-board_service	0.352047	0.066704	-0.032559	0.121149	
leg_room_service	0.304928	0.083762	-0.030316	0.136136	
baggage_handling	0.260347	-0.015452	0.019768	0.118489	
checkin_service	0.266179	0.031845	0.002456	0.041214	
cleanliness	0.259330	-0.018271	0.009432	0.108475	
online_boarding	0.338147	0.037973	0.009604	0.130396	
departure_delay_in_minutes	-0.073909	-0.009041	0.112555	-0.023817	
arrival_delay_in_minutes	-0.080487	-0.011206	0.109733	-0.025463	
customer_type_disloyal Customer	-0.292592	-0.284172	0.018938	-0.042845	
type_of_travel_Personal Travel	-0.109239	-0.044808	-0.122762	0.016839	
class_Eco	-0.278062	-0.132597	-0.138107	0.023528	
class_Eco Plus	-0.067537	-0.012922	-0.052151	0.019401	

23 rows × 23 columns

```
fig, ax = plt.subplots(figsize = (17, 8))
sns.heatmap(corr_mat, annot = True)
```

```
plt.tight_layout()
plt.show()
```



```
df.head()
```

	satisfaction	age	flight_distance	seat_comfort	departure/arrival_time_convenient	food_and_drink
0	1	65	265	0	0	0
1	1	47	2464	0	0	0
2	1	15	2138	0	0	0
3	1	60	623	0	0	0
4	1	70	354	0	0	0

5 rows × 7 columns

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report, precision_score,
```

```
X = df.drop('satisfaction', axis = 1)
y = df['satisfaction']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)
```

```
X_train.shape, X_test.shape, y_train.shape, y_test.shape
```

```
((103904, 22), (25976, 22), (103904,), (25976,))
```

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
lr = LogisticRegression(random_state = 42)
lr.fit(X_train_scaled, y_train)
y_pred = lr.predict(X_test_scaled)
```

```
y_test = y_test.astype('int')
y_pred = y_pred.astype('int')
```

```
accuracy_score(y_test, y_pred)
```

```
0.8273021250384971
```

```
confusion_mat = confusion_matrix(y_test, y_pred)
confusion_mat
```

```
array([[ 9435,  2240],
       [ 2246, 12055]])
```

```
2246 + 12055
```

```
14301
```

```
X_test.shape
```

```
(25976, 22)
```

```
y_test.value_counts()
```

	count
satisfaction	
1	14301
0	11675

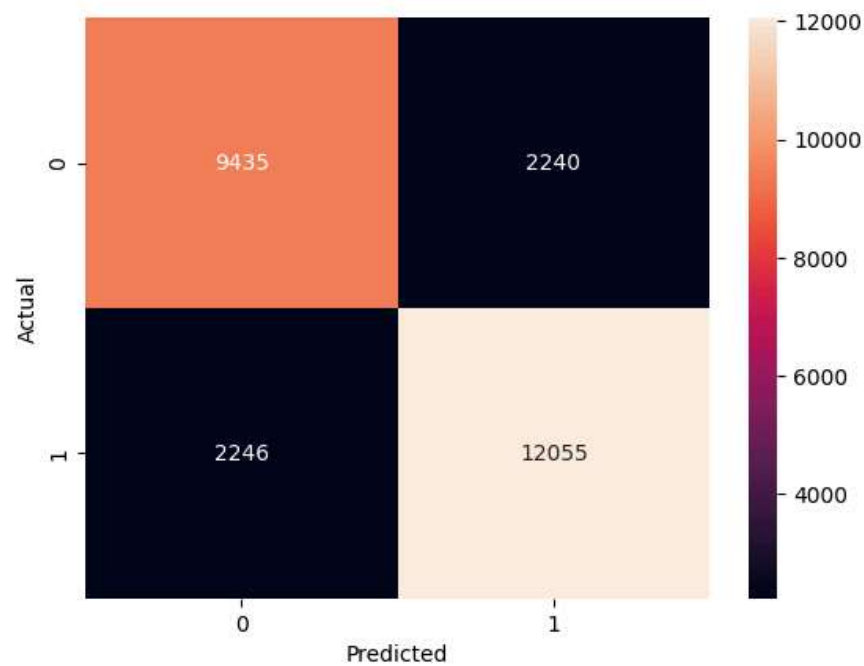
dtype: int64

```
df['satisfaction'].value_counts()
```

	count
satisfaction	
1	71087
0	58793

dtype: int64

```
sns.heatmap(confusion_mat, annot = True, fmt = 'd')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```



```
report = classification_report(y_test, y_pred, output_dict = True)
report_df = pd.DataFrame(report).transpose()
report_df.round(4)
```

	precision	recall	f1-score	support	
0	0.8077	0.8081	0.8079	11675.0000	
1	0.8433	0.8429	0.8431	14301.0000	
accuracy	0.8273	0.8273	0.8273	0.8273	
macro avg	0.8255	0.8255	0.8255	25976.0000	
weighted avg	0.8273	0.8273	0.8273	25976.0000	

```
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))
```

```
Precision: 0.8433018537950332
Recall: 0.8429480455912174
F1 Score: 0.8431249125751853
```

```
report_df.loc[['0', '1'], ['precision', 'recall', 'f1-score']]
```

	precision	recall	f1-score	
0	0.807722	0.808137	0.807929	
1	0.843302	0.842948	0.843125	

```
report_df.loc['1']
```

```

1
precision    0.843302
recall       0.842948
f1-score     0.843125
support      14301.000000

```

dtype: float64

lr.coef_

```

array([[ -0.13580725, -0.1881662 ,  0.40241355, -0.34406031, -0.30609719,
         0.17565247, -0.11871154,  0.9765042 ,  0.13688046,  0.32101372,
         0.39973266,  0.30415916,  0.10890444,  0.35578244,  0.06977573,
         0.185137 ,  0.07884574, -0.26635596, -0.73253405, -0.35633085,
        -0.35358203, -0.20314573]])

```

```

feature_importance = pd.DataFrame({'Feature': X.columns, 'Coefficient': lr.coef_[0]*100})
feature_importance.sort_values(by='Coefficient', ascending=False)

```

	Feature	Coefficient
7	inflight_entertainment	97.650420
2	seat_comfort	40.241355
10	on-board_service	39.973266
13	checkin_service	35.578244
9	ease_of_online_booking	32.101372
11	leg_room_service	30.415916
15	online_boarding	18.513700
5	gate_location	17.565247
8	online_support	13.688046
12	baggage_handling	10.890444
16	departure_delay_in_minutes	7.884574
14	cleanliness	6.977573
6	inflight_wifi_service	-11.871154
0	age	-13.580725
1	flight_distance	-18.816620
21	class_Eco Plus	-20.314573
17	arrival_delay_in_minutes	-26.635596
4	food_and_drink	-30.609719
3	departure/arrival_time_convenient	-34.406031
20	class_Eco	-35.358203
19	type_of_travel_Personal Travel	-35.633085
18	customer_type_disloyal Customer	-73.253405

```
feature_importance['Positive'] = feature_importance['Coefficient'] > 0
```

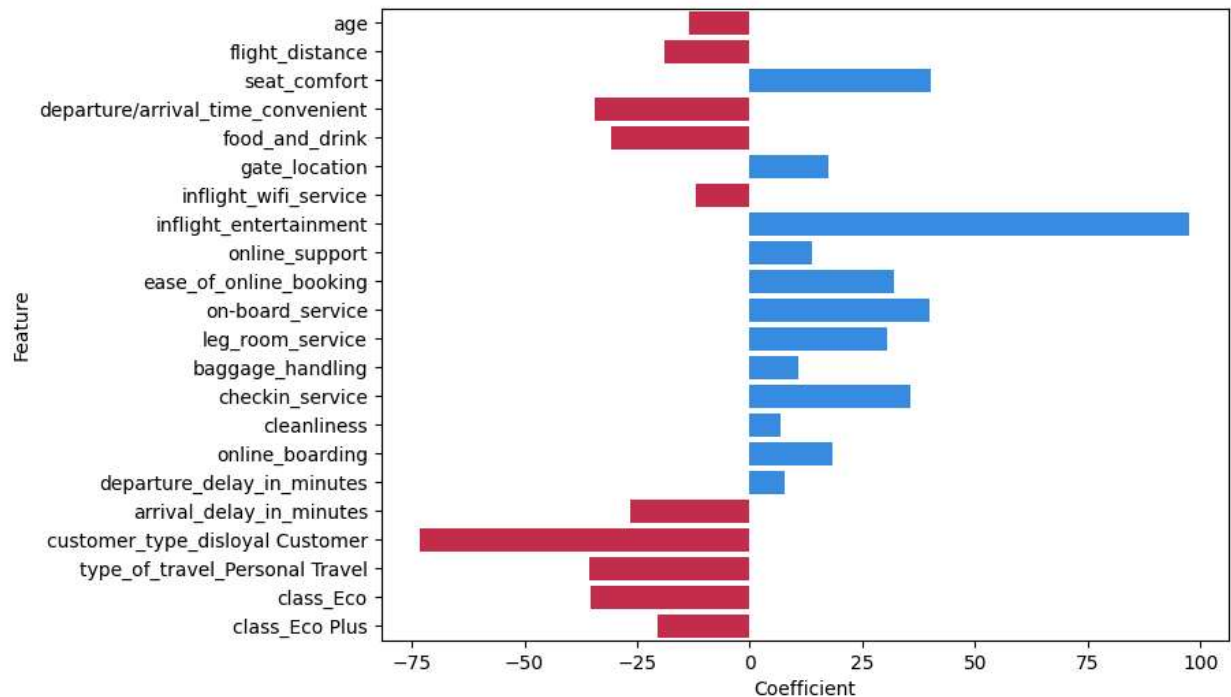
```

plt.figure(figsize=(8, 6))
sns.barplot(x='Coefficient', y='Feature',
            data=feature_importance, hue='Positive',
            palette={True: 'dodgerblue', False: 'crimson'}, legend=False)

```

```
plt.xlabel('Coefficient')
```

```
plt.xlabel('Coefficient')
plt.ylabel('Feature')
plt.show()
```



```
top_features = feature_importance.sort_values(by='Coefficient', ascending=False).head(12)
top_features
```

	Feature	Coefficient	
7	inflight_entertainment	97.650420	
2	seat_comfort	40.241355	
10	on-board_service	39.973266	
13	checkin_service	35.578244	
9	ease_of_online_booking	32.101372	
11	leg_room_service	30.415916	
15	online_boarding	18.513700	
5	gate_location	17.565247	
8	online_support	13.688046	
12	baggage_handling	10.890444	
16	departure_delay_in_minutes	7.884574	
14	cleanliness	6.977573	

Next steps: [Generate code with top_features](#) [New interactive sheet](#)

Features

The model was fitted and achieved 82.73% accuracy on the test set.

Confusion Matrix

	Predicted: 0 (Dissatisfied)	Predicted: 1 (Satisfied)
Actual: 0	9435	2240

	Predicted: 0 (Dissatisfied)	Predicted: 1 (Satisfied)
Actual: 1	2246	12055

The model correctly classified 9,435 as Dissatisfied out of 1,167 Dissatisfied passengers and 12,055 as Satisfied out of 14,301 Satisfied passengers.

The model achieves a Precision: 84.33, Recall: 84.29%, F1 Score: 84.31%. This means that 84.33% of the time, the model correctly predict the postive class (satisfied), recall indicates that 84% of the times the model correctly identify the positive class and indicates a strong balance between catching positives values and avoiding false alarms.

Model Feature Coefficients

The table below shows the impact of each feature on the model prediction. Positive values increase the likelihood of the positive class, while negative values decrease it.

Feature	Coefficient
inflight_entertainment	97.650420
seat_comfort	40.241355
on-board_service	39.973266
checkin_service	35.578244
ease_of_online_booking	32.101372
leg_room_service	30.415916
online_boarding	18.513700
gate_location	17.565247
online_support	13.688046
baggage_handling	10.890444
departure_delay_in_minutes	7.884574
cleanliness	6.977573
inflight_wifi_service	-11.871154
age	-13.580725
flight_distance	-18.816620
class_Eco Plus	-20.314573
arrival_delay_in_minutes	-26.635596
food_and_drink	-30.609719
departure/arrival_time_convenient	-34.406031
class_Eco	-35.358203
type_of_travel_Personal Travel	-35.633085
customer_type_disloyal Customer	-73.253405

Feature Importance Analysis and Business Recommendations

The model results indicate that customer satisfaction is primarily driven by the quality of the onboard experience. Among all features, **inflight entertainment** has the strongest positive influence on satisfaction, followed by **seat comfort**, **on-board service**, **check-in service**, **ease of online booking**, and **leg room service**. This suggests that passengers place significant value on comfort, convenience, and service quality throughout their journey. Digital touchpoints such as online booking, online boarding, and customer support also contribute positively, highlighting the importance of a seamless customer experience before and during travel.

The strongest negative factor is being a **disloyal customer**, indicating that customers without an established relationship with the airline are substantially less likely to report satisfaction. Other notable negative factors include **personal travel**, **economy class travel**, **arrival delays**, **food and drink quality**, and the convenience of departure and arrival times. These findings suggest that leisure travelers and economy-class passengers generally have lower satisfaction levels, while operational inefficiencies and service shortcomings negatively impact the customer experience.

Business Recommendations

To improve overall customer satisfaction, the airline should prioritize investments in onboard experience, particularly inflight entertainment, seating comfort, cabin service, and digital customer interactions. Enhancing loyalty programs and retention initiatives could help convert disloyal customers into repeat passengers, thereby improving satisfaction levels. Additionally, reducing arrival delays through operational improvements should remain a key focus, as delays significantly diminish customer experience. Finally, targeted improvements in food and beverage offerings and tailored service enhancements for economy-class and personal-travel customers may help address satisfaction gaps within these customer segments.

Business Insights and Recommendations

Key Findings

- Improving **inflight entertainment** is likely to have the greatest impact on customer satisfaction, making it the most influential service-related factor in the model.
- Enhancing **seat comfort**, **on-board service**, and **check-in service** could significantly improve the overall passenger experience and increase satisfaction levels.
- Streamlining the **online booking** and **online boarding** processes may improve customer perceptions by reducing friction throughout the travel journey.
- Better **leg room availability** and improved **gate location accessibility** are associated with higher customer satisfaction.
- Efficient **online support** and **baggage handling** contribute positively to customer experience and should be maintained as operational strengths.
- **Disloyal customers** are substantially less likely to be satisfied than loyal customers, indicating a strong relationship between customer retention and satisfaction.
- Passengers traveling in **Economy** and **Economy Plus** classes tend to report lower satisfaction levels compared to premium cabin passengers.
- **Personal travel** customers appear less satisfied than business travelers, suggesting differing expectations and service needs across traveler segments.
- Increased **arrival delays** are strongly associated with lower customer satisfaction, highlighting the importance of operational reliability.
- Lower ratings for **food and drink quality**, **departure/arrival time convenience**, and **inflight Wi-Fi service** are linked to reduced customer satisfaction.

Business Recommendations

- Invest in upgrading and expanding **inflight entertainment systems**, as improvements in this area are likely to generate the largest increase in customer satisfaction.
- Enhance **seat comfort and cabin service standards** through staff training and service quality initiatives.
- Simplify the **online booking and boarding experience** by reducing process complexity and improving digital platform usability.
- Develop targeted **customer loyalty programs** to encourage repeat patronage and improve satisfaction among currently disloyal customers.
- Implement operational strategies aimed at reducing **arrival delays**, such as schedule optimization and proactive disruption management.
- Improve **food and beverage offerings**, particularly on routes where customer satisfaction scores are lower.
- Upgrade **inflight Wi-Fi reliability and speed** to better meet passenger expectations, especially for travelers who require connectivity during flights.
- Introduce targeted service improvements for **Economy-class passengers**, who represent the segment with the lowest satisfaction levels.
- Create tailored travel packages and customer engagement initiatives for **personal travelers** to address their unique needs and expectations.
- Continuously monitor customer feedback across key service touchpoints to identify emerging satisfaction drivers and areas requiring improvement.